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Multi-objective Pareto hyperparameter optimization for joint detection and localization of forged identity documents

Hernán Díaz Rodríguez[a,**]

[a]*Department of Computer Science, University of Oviedo, Asturias, Spain*

ABSTRACT

Automated forensic analysis of identity documents requires simultaneously detecting whether a document has been tampered with and identifying which regions were altered. These two objectives are structurally coupled but exist in tension: optimizing one with a fixed loss weight may degrade the other. We propose DocVerify, a multi-task convolutional network paired with a multi-objective hyperparameter optimization (HPO) framework that treats detection quality (PR-AUC) and localization quality (Dice) as simultaneous objectives. Using the Multiobjective Tree-structured Parzen Estimator (MOTPE) within a 10-fold nested cross-validation protocol (50 trials per outer fold, 500 unique configurations evaluated across 5 inner splits, totaling 2 500 individual model trainings), DocVerify selects configurations from the Pareto-optimal front by minimizing Euclidean distance to the ideal point (1, 1), without any manual weighting. Evaluated on the FantasyID benchmark, DocVerify achieves PR-AUC = 0.9921 ± 0.0058 and Dice = 0.856 ± 0.030 under nested cross-validation, and detection F1@0.5 = 0.969 ± 0.014 with localization F1 = 0.807 ± 0.096 on the blind test set (30 seeds). An ablation confirms that Pareto-optimized loss weighting provides a statistically significant Dice improvement over equal-weight joint training ($p < 0.001$, $d = 1.01$, large effect) while maintaining equivalent detection accuracy. These figures characterize in-domain performance on FantasyID; zero-shot transfer to the out-of-domain SIDTD benchmark falls to near chance (PR-AUC 0.555), so the reported accuracy should be read as bounded to the training domain.

1. Introduction

Identity document fraud poses a growing threat to national security, financial integrity, and border control worldwide. The widespread availability of image-editing software has made it possible to produce convincing forgeries of passports, national identity cards, and driver's licenses at low cost. Automated forensic analysis of identity documents is therefore a critical component of modern digital identity verification pipelines.

Detection of forged identity documents is inherently a two-level problem. At the *document level*, a system must decide whether a presented document is genuine or has been tampered with (binary detection). At the *pixel level*, it must also identify and delineate which specific regions have been altered (forgery localization). Both objectives are operationally necessary.

Binary detection alone is insufficient for real-world forensic deployments. A positive detection decision provides no *actionable information* to the investigator: it does not reveal whether the fraudster altered the portrait photograph, falsified textual fields, or combined multiple manipulations. Without spatial evidence, a human expert cannot determine the nature of the fraud or build a legal case. Detection systems are also increasingly subject to *explainability requirements* under emerging AI regulations: a binary "tampered" label cannot satisfy the obligation to provide a meaningful explanation of the automated decision. Conversely, localization alone does not yield a reliable document-level confidence score suitable for automated accept/reject decisions. A practical forensic system must address both tasks simultaneously.

The primary technical difficulty in joint detection and localization is that the two objectives are in *structural tension*: the classification head benefits from global image features, while

**Corresponding author.
e-mail: diazhernan@uniovi.es (Hernán Díaz Rodríguez)

the segmentation head requires spatially precise, pixel-level representations. Prior work either treats the two tasks independently [1] or collapses both losses into a scalar objective via a fixed weight λ [2]. The choice of λ is arbitrary and dataset-dependent. A second challenge is the risk of over-fitting the model-selection criterion: hyperparameter optimization (HPO) that maximizes PR-AUC may sacrifice Dice, and vice versa.

This paper introduces DocVerify, a multi-task forensic document analysis system that addresses both challenges. Our contributions are:

1. *Multi-objective Pareto HPO for multitask forensics.* Both PR-AUC and Dice are treated as simultaneous optimization objectives. The optimal configuration is selected from the Pareto front by minimizing the Euclidean distance to the ideal point $(1,\ 1)$, requiring no manual task-weight tuning.
2. *DocVerify architecture.* An end-to-end multi-task CNN comprising a lightweight convolutional encoder coupled to a U-Net decoder for pixel-level mask prediction and a classification head for document-level detection, trained jointly with a combined BCE+Dice segmentation loss.
3. *Rigorous nested cross-validation.* A 10-fold outer $\times$ 5-fold inner nested cross-validation scheme with 50 Multiobjective Tree-structured Parzen Estimator (MOTPE) trials per outer fold (2 500 HPO evaluations total). Final performance is estimated on a blind holdout using 30 independent seeds, with Wilcoxon signed-rank tests [3] and Holm–Bonferroni correction [4].

The source code and experimental artifacts are publicly available at `https://github.com/HernanDiaz/IDVerify/`.

The remainder of the paper is organized as follows. Section 2 reviews related work. Section 3 describes the DocVerify architecture and HPO framework. Section 4 presents experimental results. Section 5 discusses the findings and limitations. Section 6 concludes.

2. Related Work

2.1. Document and Image Forgery Detection

Deep learning has substantially advanced general-purpose image forensics: CNNs can detect subtle statistical fingerprints left by editing operations, often outperforming hand-crafted pipelines [5]. Cozzolino and Verdoliva [6] (Noiseprint) train a CNN via Siamese contrastive learning to extract camera-model noise residuals; forgeries appear as anomalies in this residual without requiring semantic understanding. Guillaro et al. [1] (TruFor) generalize this with a transformer fusion network (NoisePrint++) that jointly models RGB semantics and noise-pattern anomalies, achieving strong performance on general image forensics benchmarks with ~828 000 pre-training images. Closely related, deepfake and face-manipulation detection has advanced rapidly in this journal: Liang et al. [7] pair a two-stream network with hierarchical supervision, Leporoni et al. [8] fuse RGB and depth cues, and Wang et al. [9] add a self-attention discriminator, each improving robustness to synthetic facial forgeries.

Identity document forgery introduces unique challenges: forged regions are often small (a single text field or portrait), the structured background (guilloché patterns, UV-reactive inks) creates strong false negatives for general-purpose noise detectors, and document type diversity demands generalization beyond natural-image pre-training. Official evaluations on FantasyID confirm that general-purpose baselines (TruFor, MMFusion [10], FatFormer [11], UniFD [12]) all achieve false-negative rates above 50% at a 10% false-positive threshold [13], motivating domain-specific approaches.

Among systems trained exclusively on FantasyID, George and Marcel [14] propose EdgeDoc, a lightweight CNN-Transformer encoder (EdgeNeXt [15]) fed with the green channel and a NoisePrint residual, with a U-Net decoder optimized jointly for classification and localization. Yu et al. [16] introduce Re-MTKD, a reinforcement-learning multi-teacher knowledge distillation framework using a ConvNeXt v2 encoder; the Sunlight team adapted this to identity documents using over 60 000 additional external images, winning both tracks of the DeepID 2025 Challenge [17].

DocVerify differs from all prior document-forensics systems in three respects: (i) it systematically searches the task-weight λ_{mask} via multi-objective HPO rather than fixing it heuristically; (ii) it selects the final configuration via a Pareto-optimal criterion; and (iii) it reports uncertainty through nested cross-validation, providing statistically rigorous performance estimates.

2.2. Multi-Task Learning and HPO

In multi-task learning, the dominant approach combines task losses via scalar scalarization [18] with fixed weights set before training. Cipolla et al. [2] propose learning these weights via homoscedastic uncertainty; Liu et al. [19] introduce gradient normalization. All scalar approaches commit to a fixed trade-off before training begins.

Multi-objective HPO extends single-objective search to jointly optimize multiple metrics [20]. The Multiobjective TPE (MOTPE) [21] adapts the TPE surrogate [22] to the multi-objective setting, achieving competitive Pareto-front coverage with far fewer function evaluations than evolutionary methods such as NSGA-II [23], which is critical when each evaluation requires training a deep network. To our knowledge, this is the first work to apply Pareto-based multi-objective HPO to joint identity-document forgery detection and localization within a nested cross-validation framework.

3. Methodology

3.1. Dataset

We train and evaluate DocVerify on FantasyID [13], a public benchmark for ID-document forgery detection and localization from Idiap for the DeepID 2025 Challenge [17]. The FantasyID release we process contains 3 284 labeled images of synthetic fantasy identity documents: *bonafide* (genuine) and *attack* (tampered via face swap or text inpainting).

Each image is accompanied by JSON annotations specifying bounding-box regions with `region_provenance: altered`.

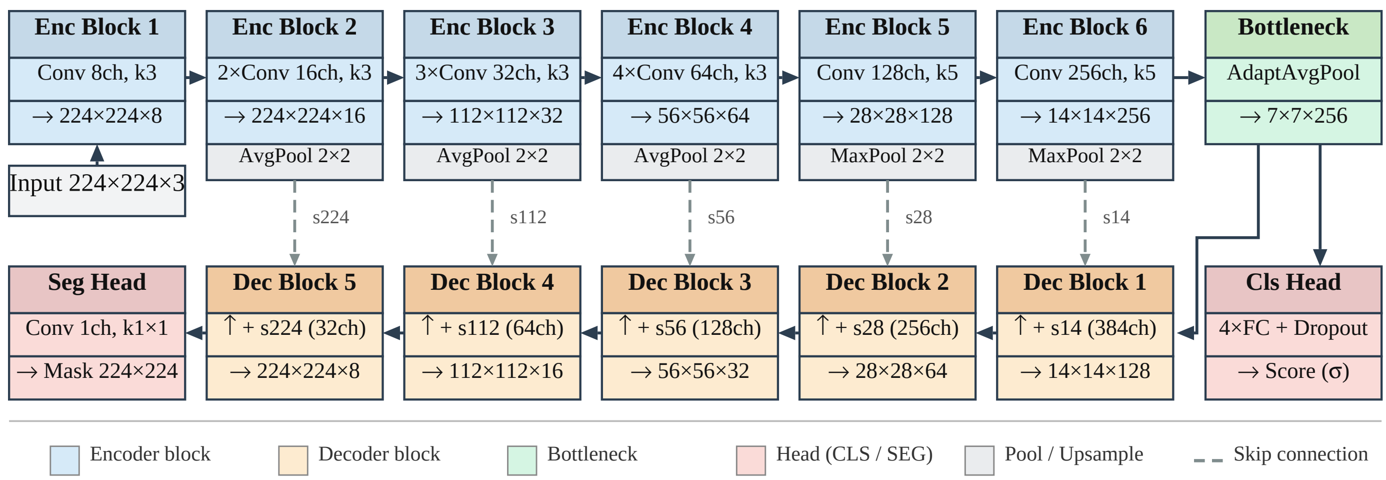

Fig. 1. DocVerify architecture. The encoder maps a 3×224×224 input to a 256×7×7 bottleneck through six convolutional blocks (8→256 channels). The U-Net decoder reconstructs the forgery mask via skip connections; the classification head outputs a document-level tampered/genuine score.

We convert these to binary segmentation masks at training resolution (224 × 224 pixels). We apply `GroupShuffleSplit` to all 3 284 available FantasyID images (grouping by document identity, test_size = 0.15, `random_state=42`) to obtain approximately 2 791 development images and 493 blind holdout images. This split was performed once, before any hyperparameter search or model-design decision, and the holdout was accessed only for the final blind evaluation (Section 4.4). The dataset is class-imbalanced: 2 351 attack images (71.6%) and 933 bonafide images (28.4%); this imbalance is preserved in all splits and not corrected via resampling or loss weighting. The mild imbalance (2.5:1, attack-majority) leaves PR-AUC robust; a `pos_weight` sweep confirms no significant effect.

3.2. Architecture

DocVerify is a multi-task encoder-decoder network for simultaneous document-level classification and pixel-level forgery localization (Fig. 1).

3.2.1. Encoder

The encoder is a six-block lightweight CNN inspired by Patel et al. [24], a deepfake-detection architecture specifically designed for forgery artifact extraction. A lightweight encoder is a deliberate design choice: the nested CV protocol requires training ~2 500 models, so per-trial cost must be kept low. The channel progression is 8 → 16 → 32 → 64 → 128 → 256 filters. All layers use LeakyReLU ($\alpha = 0.2$) and batch normalization. Blocks 1–4 use 3×3 convolutions; blocks 5–6 use 5×5 convolutions. Downsampling uses average pooling after blocks 2 and 3 (to preserve boundary information) and max pooling after blocks 4 and 5 (for translation invariance).

3.2.2. Classification Head

Global average pooling is applied to the 7×7 bottleneck, yielding a 256-dimensional vector passed through four fully-connected layers (256→32→16→16→1). Dropout (rate p) precedes each linear layer and LeakyReLU ($\alpha = 0.2$) each hidden layer; the final layer is linear, producing a classification logit $\hat{y}_{\text{cls}}$.

3.2.3. Segmentation Head (U-Net Decoder)

A five-stage U-Net decoder [25] reconstructs the spatial mask. Each stage applies bilinear ×2 upsampling, concatenates channel-wise the matching-resolution encoder skip (tapped before pooling), then two 3×3 convolutions with LeakyReLU ($\alpha = 0.2$) and no batch normalization. The channel width halves at each stage from C_{dec} (a tunable HPO parameter) down to $C_{\text{dec}}/16$. A final 1×1 convolution produces the segmentation logit $\hat{Y}_{\text{mask}}$.

3.3. Loss Functions

The combined training loss is:

$$\mathcal{L} = \mathcal{L}_{\text{cls}} + \lambda_{\text{mask}}\, \mathcal{L}_{\text{seg}}, \tag{1}$$

where, in Eq. 1, $\lambda_{\text{mask}} \in [0.5, 3.0]$ is included in the HPO search space. $\mathcal{L}_{\text{cls}} = \text{BCEWithLogits}(\hat{y}_{\text{cls}}, y_{\text{cls}})$. The segmentation loss combines BCE and Dice:

$$\mathcal{L}_{\text{seg}} = \text{BCE}(\hat{Y}, Y) + 1 - \frac{2 \sum_i p_i g_i + \epsilon}{\sum_i p_i + \sum_i g_i + \epsilon}, \tag{2}$$

The Dice term in Eq. 2, as introduced for volumetric segmentation [26], addresses foreground/background imbalance in small forgery regions.

All models are trained with AdamW and ReduceLROnPlateau (factor 0.5, patience 3). Mixed-precision training (FP16/FP32 via PyTorch AMP) is used throughout.

3.4. Multi-Objective HPO

We optimize two complementary metrics: *PR-AUC* (area under the precision-recall curve, measuring document-level detection quality) and *Dice* ($2|P \cap G|/(|P| + |G|)$, measuring pixel-level mask overlap). Our HPO formulation simultaneously maximizes:

$$\max_{\boldsymbol{\theta} \in \Theta} \left(\text{PR-AUC}(\boldsymbol{\theta}),\ \text{Dice}(\boldsymbol{\theta})\right). \tag{3}$$

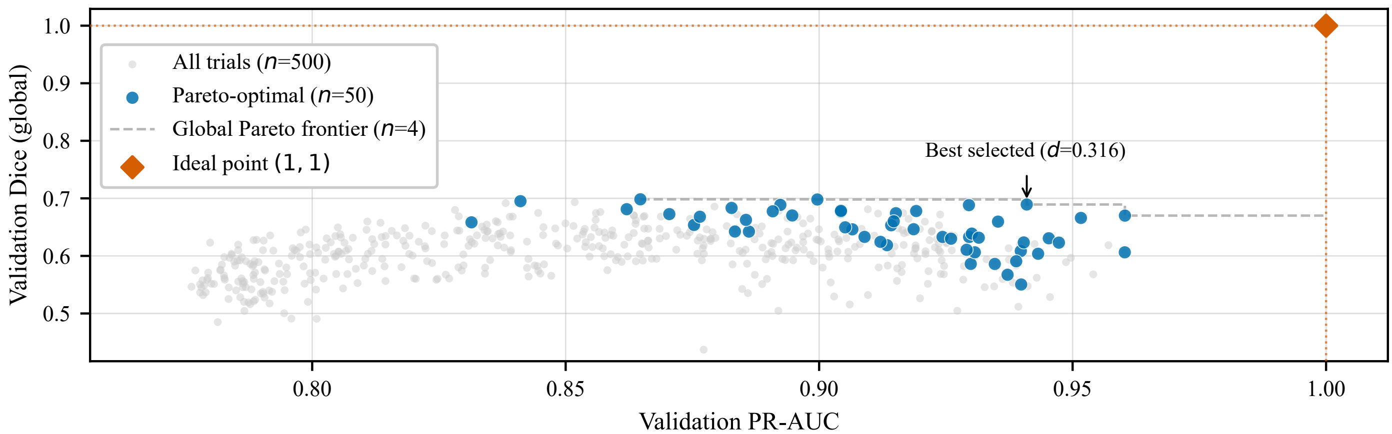

Fig. 2. Validation objective space across all 500 HPO trials (10 folds × 50 MOTPE trials). Grey: evaluated; blue: Pareto-optimal; dashed: Pareto frontier. Red diamond: ideal point $(1, 1)$**; annotated point: selected configuration (distance** $d = 0.316$**).**

Table 1. HPO search space and final selected configuration (Section 4.3). All log-uniform except dropout and λ_{mask} (uniform) and decoder channels.

Parameter	Range	Selected
η (learning rate)	$[5\times10^{-5},\ 9\times10^{-4}]$	5.21×10^{-4}
λ_{wd} (weight decay)	$[10^{-7},\ 10^{-4}]$	4.27×10^{-5}
p (dropout)	$[0.1,\ 0.4]$	0.103
C_{dec} (decoder channels)	$\{96, 128, 192, 256\}$	192
λ_{mask} (mask loss weight)	$[0.5,\ 3.0]$	2.46

In Eq. 3, Θ is the search space defined in Table 1.

We use MOTPE [21] as implemented in Optuna [27]. After 50 trials on the inner splits, we select the configuration $\boldsymbol{\theta}^*$ from the Pareto front $\mathcal{P}$ by minimizing the Euclidean distance to the ideal point (Eq. 4):

$$\boldsymbol{\theta}^* = \arg\min_{\boldsymbol{\theta}\in\mathcal{P}} \sqrt{(1-\text{PR-AUC})^2 + (1-\text{Dice})^2}. \tag{4}$$

This criterion requires no user-specified weighting and assigns equal importance to detection and localization. Figure 2 illustrates the resulting objective space.

3.5. Nested Cross-Validation

To obtain unbiased estimates while keeping HPO blind to test data, we use a three-level nested CV scheme [28]. The outer loop uses 10 stratified folds of the training set (85%); each fold serves once as the test set. For each outer fold, the remaining data is split into 5 inner folds on which 50 MOTPE trials are run (up to 15 epochs each, patience = 1). The optimal $\boldsymbol{\theta}^*$ is then used to retrain from scratch on all outer-fold training data (up to 100 epochs, early stopping patience = 12 on the validation distance to the ideal point (Eq. 4), as in Pareto selection). This design ensures HPO never sees test data: 50 MOTPE trials per outer fold yield 500 unique hyperparameter configurations in total (10 outer folds × 50 trials); because each configuration is evaluated on 5 inner splits, this corresponds to $10\times50\times5 = 2\,500$ individual model trainings, plus 10 final models retrained on full outer-fold data.

4. Experiments and Results

4.1. Experimental Setup

All experiments are conducted on a single workstation running Windows 11, equipped with an AMD Ryzen 5 3400G (3.70 GHz), 32 GB DDR4-3200 RAM, and an NVIDIA RTX 5060 Ti (16 GB VRAM), using PyTorch 2.10.0 with CUDA 13.0 and Optuna 3.x. Training uses a batch size of 64 and no data augmentation (standard augmentation, rotation, brightness, contrast, blur and JPEG, significantly degraded every metric on the FantasyID holdout; see [29]). All images are resized to 224×224 pixels and normalised to $[0, 1]$ prior to caching in GPU memory. The full NCV pipeline required approximately 59 hours; VRAM caching reduces per-trial training time from ~12 to ~3.5 minutes. This 59-hour budget covers the full scientific protocol; retraining the single deployed model from scratch takes 7.7 ± 1.0 min.

4.2. Nested Cross-Validation Results

Table 2 reports per-fold test metrics. PR-AUC is consistently high across all folds (0.977–0.999, mean 0.9921 ± 0.0058). Dice shows slightly more fold-to-fold variability (0.800–0.909, mean 0.856 ± 0.030), reflecting the difficulty of pixel-level localization. The Pareto-selected λ_{mask} clusters between 2.3 and 2.7 in eight of ten folds; the two outlier folds (F5, F9) correspond to the two lowest Dice scores, confirming that λ_{mask} is a decisive driver of localization quality.

4.3. Final Model Configuration

The cross-validation above yields ten fold-specific configurations and estimates generalization, but the deployed system is a single model. We select the configuration with the smallest inner-CV distance to the ideal point $(1, 1)$ across all outer folds (Eq. 4; Table 1), which fixes both the architecture (decoder width) and the Pareto-optimized loss weight $\lambda_{\text{mask}} = 2.46$, consistent with the 2.3–2.7 cluster of Section 4.2. This final configuration is retrained from scratch on the development set (up to 100 epochs with early stopping, as in Section 3.5) and evaluated on the held-out 15% blind-test set; we repeat the retraining over 30 independent seeds to quantify run-to-run variability.

Table 2. Per-fold test results from 10-fold NCV. $\overline{w}_{\text{mask}}$: selected loss weight.

Fold	PR-AUC	Dice	$\overline{w}_{\text{mask}}$
F1	0.9939	0.861	2.46
F2	0.9769	0.871	2.70
F3	0.9993	0.909	2.47
F4	0.9915	0.858	2.43
F5	0.9948	0.827	1.41
F6	0.9936	0.874	2.89
F7	0.9940	0.861	2.67
F8	0.9967	0.887	2.29
F9	0.9896	0.807	1.95
F10	0.9909	0.800	2.65
Mean (Std)	**0.9921 (0.0058)**	**0.856 (0.030)**	**2.39 (0.43)**

4.4. Ablation and Selection-Criterion Study

To isolate each design decision, we evaluate this final model against five variants under the same 30-seed blind-test protocol (Table 3). Three variants probe the architecture and loss weighting: `cls_only` (classification head only); `seg_only` (segmentation head only, detection by thresholding mean mask probability); and `unweighted_losses` (full multitask, $\lambda_{\text{cls}} = \lambda_{\text{mask}} = 0.5$, bypassing HPO). Two further variants probe the HPO selection criterion applied to the *same* 500 MOTPE trials: `by_prauc` (the trial maximizing validation PR-AUC) and `by_dice` (the trial maximizing validation Dice), against our Pareto criterion (distance to the ideal point).

Regarding the architecture and loss variants, three findings emerge. First, joint multitask training is essential: `cls_only` produces near-zero Dice (0.002 ± 0.008). Second, a dedicated classification head is necessary: `seg_only` achieves competitive Dice (0.867) but PR-AUC drops to 0.728 ± 0.053, confirming that the segmentation logit alone is an unreliable detection signal. Third, Pareto-optimized weights outperform equal weights specifically on localization: `unweighted_losses` matches the full model on PR-AUC ($p = 0.083$) but shows significantly lower Dice ($p = 1.3 \times 10^{-4}$, $d = 1.01$, large effect), supporting the HPO contribution.

A key question for any multi-objective HPO framework is whether the added complexity over simpler scalar alternatives yields measurable gains. The advantage over scalar selection is real but partial and metric-specific rather than uniform. Against `by_dice`, Pareto is significantly better after Holm correction on four of five metrics: PR-AUC (+0.0019, $d = 0.64$, $p = 0.007$), Dice (+0.018, $d = 1.00$, $p < 0.001$), balanced accuracy ($d = 0.45$, $p = 0.040$) and F1-macro ($d = 0.47$, $p = 0.043$), though only the Dice gain reaches a large effect. Against `by_prauc` the advantage is narrower: only Dice ($d = 0.59$, $p = 0.043$) and F1-macro ($d = 0.46$, $p = 0.050$, at the significance threshold) are significant, while PR-AUC, balanced accuracy and F1-attack are not ($p > 0.06$). Effect sizes are otherwise small-to-medium ($d \approx 0.4$–0.65) and the absolute gains modest ($\Delta \lesssim 0.02$), so the practical benefit is concentrated in localization quality; its broader value is that distance-to-ideal selection attains this without committing a priori to a primary objective (full per-metric statistics in Supplementary Table S4).

Beyond predictive accuracy, Pareto selection requires no a priori commitment to a primary objective, making the full trade-off landscape observable (Fig. 2) and remaining structurally consistent with the DeepID challenge protocol, which ranks detection and localization independently.

4.5. Comparison with Related Systems

DocVerify was developed after the DeepID 2025 Challenge [17] deadline; we report results on our internal 15% hold-out (30 seeds), which differs from the official challenge set, so any comparison is indicative only. Training data also differ markedly. The leading entries rely on external data: Sunlight (1st) augmented FantasyID with over 60 000 identity images, while the third-place EdgeDoc+TruFor fusion and the TruFor baseline [1][1] build on a backbone pre-trained on ~828 000 manipulation images. EdgeDoc standalone [14], UAM-Biometrics, and DocVerify instead train on FantasyID alone. With these caveats, DocVerify reaches detection F1@0.5 (the F1 score, $2PR/(P+R)$, of the bonafide/attack decision at classification threshold 0.5) = 0.969 ± 0.014 and localization F1_{loc} (pixel-level F1 of the predicted mask) = 0.807 ± 0.096. Reported challenge figures span EdgeDoc standalone (0.43 [14]), UAM-Biometrics (0.712), the EdgeDoc+TruFor fusion (0.958 [17], on the pre-trained TruFor backbone) and the winner Sunlight (0.991, F1_{loc} 0.784). As evaluation sets and training data differ, we make no superiority claim and imply no ranking; the figures only add context.

4.6. Generalization

Two experiments probe whether the approach generalizes beyond its specific setting. *Across architectures*, we re-ran the identical MOTPE protocol (50 trials, 5 inner splits, Pareto selection, 30-seed blind test; a single study rather than the full nested loop, to bound cost) with the encoder replaced by an ImageNet-pretrained ResNet-18 feeding the same U-Net decoder. MOTPE selected $\lambda_{\text{mask}} = 2.69$ (vs. 2.46 for DocVerify), so the Pareto balance is consistent across backbones. It reaches PR-AUC 0.994 ± 0.003 and Dice 0.889 ± 0.011 (30 seeds); paired Wilcoxon + Holm gives DocVerify a small PR-AUC edge ($d = 0.75$) and ResNet-18 a Dice/F1 edge ($d = 0.60$–0.67), with balanced accuracy tied. Neither dominates, indicating the selection methodology is not tied to a specific backbone [29]. *Across datasets*, we evaluate the FantasyID blind-test models zero-shot on the out-of-domain SIDTD dataset [31], using only its *templates* subset (2 222 images: 1 000 bonafide, 1 222 attack) rather than the full SIDTD video collection. Zero-shot performance drops to near chance (PR-AUC 0.555 ± 0.011, ROC-AUC 0.504 ± 0.009): the FantasyID→SIDTD domain gap is too wide for a model trained on FantasyID alone. Retraining the same architecture on an 80/20 stratified split of the templates subset (1 777 train, 445 test) recovers PR-AUC 0.878 ± 0.016, ROC-AUC 0.826 ± 0.023, whereas a few-shot sweep (10–400 images)

[1] For completeness, per-seed scores for a TruFor model fine-tuned on FantasyID are released with the code (`trufor_finetuned_scores.csv`); we omit them from the comparison here because, like the challenge figures, they were not obtained on our internal holdout, so no controlled comparison is possible.

Table 3. Ablation and selection-criterion comparison (blind test, 30 seeds); all variants vs. Pareto (ours). p: paired Wilcoxon [3] + Holm [4] (significant at $p < 0.05$); d: Cohen's d [30] ($|d| \approx 0.2$ small, 0.5 medium, ≥ 0.8 large).

	PR-AUC			Dice		
Variant	Mean ± Std	p	d	Mean ± Std	p	d
Pareto (ours)	**0.9967 ± 0.0021**	—	—	**0.875 ± 0.018**	—	—
`by_prauc`	0.9953 ± 0.0021	0.06	0.49	0.859 ± 0.023	0.04	0.59
`by_dice`	0.9948 ± 0.0027	0.007	0.64	0.857 ± 0.015	< 0.001	1.00
`cls_only`	0.9503 ± 0.0744	< 0.001	0.62	0.002 ± 0.008	< 0.001	47.2
`seg_only`	0.7277 ± 0.0528	< 0.001	5.15	0.867 ± 0.023	0.29	0.32
`unweighted_losses`	0.9958 ± 0.0016	0.08	0.37	0.857 ± 0.016	< 0.001	1.01

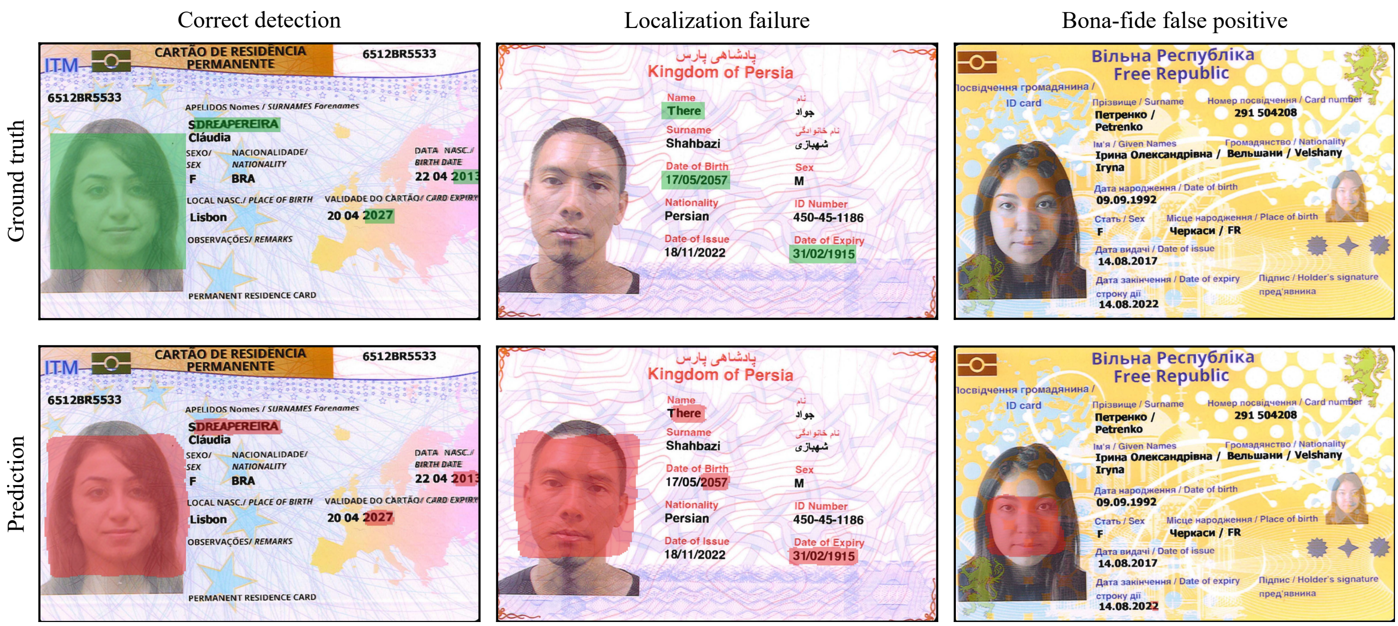

Fig. 3. Representative blind-test predictions (single seed). Top row: ground-truth altered regions (green); bottom row: regions predicted by DocVerify (red). Left: a forged photograph correctly detected and localized. Centre: a forgery correctly flagged but poorly localized. Right: a bonafide document misclassified as an attack.

stays at chance: only full in-domain training closes the gap, so it reflects domain shift, not architectural capacity. The zero-shot scope of the current FantasyID-trained system is thus bounded to its training domain.

4.7. Qualitative Analysis

Figure 3 contrasts three blind-test outcomes: a forged photograph precisely localized (left, $p_{\text{attack}} = 0.96$, Dice 0.96), and the two failure modes, a forgery correctly flagged but with its mask drifting onto the portrait (centre, $p_{\text{attack}} = 1.00$, Dice 0.22) and a bonafide document wrongly flagged with a spurious mask (right, $p_{\text{attack}} = 0.96$), illustrating the model's localization variance and false positives.

5. Discussion

5.1. Analysis of Key Findings

The ablation reveals a task asymmetry: `seg_only` achieves competitive Dice (0.867 ± 0.023, $p = 0.288$ vs. Pareto) but its PR-AUC collapses to 0.728 ± 0.053. The U-Net decoder captures forgery evidence for pixel-level prediction yet cannot aggregate it into a reliable document-level decision, confirming the shared encoder, regularized by the segmentation loss, learns spatially-aware features that improve detection.

Comparing Pareto against `unweighted_losses` isolates the HPO loss-weighting contribution: these differ only in whether λ_{mask} is HPO-set (2.46 deployed, NCV fold mean 2.39 ± 0.43) or fixed at 0.5. Pareto-optimized weights yield significantly higher Dice ($p = 1.3 \times 10^{-4}$, $d = 1.01$) at equal PR-AUC, confirming $\lambda_{\text{mask}} \approx 2.4$ is broadly optimal. The comparison against the two scalar criteria (Sec. 4.4) further shows the gain stems from the selection rule itself: scalar criteria over-commit to one axis, whereas distance-to-ideal selection balances both objectives without a priori weighting.

5.2. Limitations

Computational cost.. The full nested-CV protocol (2,500 trainings, ~59 h) is computationally heavy, but this reflects the unbiased-estimation protocol, not MOTPE itself: HPO alone can be run on a single train/validation split at a fraction of the cost, and the deployed model retrains in 7.7 ± 1.0 min.

Generalization.. Trained on FantasyID alone, DocVerify falls to near chance zero-shot on SIDTD (Sec. 4.6); even the top challenge entries generalized only by training on large external corpora (Sec. 4.5), making multi-source training the main future direction. Standard augmentation does not close this gap: augmentation-trained models remain at near chance zero-shot on SIDTD (Supplementary Table S8), indicating the gap reflects domain shift rather than limited input variability.

6. Conclusion

We presented DocVerify, a multi-task CNN for joint detection and localization of forged identity documents, paired with a multi-objective Pareto HPO framework that selects hyperparameter configurations by optimizing PR-AUC and Dice simultaneously without fixing their relative importance in advance.

Our main findings are: (i) learning detection and localization jointly benefits both tasks: a single-task model performs well only on the axis it targets (detection-only Dice 0.002; localization-only PR-AUC 0.728; both $p < 0.001$), whereas the joint model attains both; it further yields higher Dice than equal-weight training ($p = 1.3 \times 10^{-4}$, $d = 1.01$) at no PR-AUC cost, and distance-to-ideal selection improves on scalar PR-AUC- or Dice-maximizing selection chiefly in localization, a partial and metric-specific edge; (ii) under a rigorous 10-fold nested cross-validation protocol, DocVerify achieves PR-AUC = 0.9921 ± 0.0058 and Dice = 0.856 ± 0.030, demonstrating consistent generalization across data partitions; and (iii) trained on FantasyID alone, DocVerify attains F1@0.5 = 0.969 ± 0.014 and $F1_{loc}$ = 0.807 ± 0.096 on our internal holdout, reported alongside the DeepID 2025 Challenge for context only, as the evaluation sets differ.

Two strengths and two limitations frame these results. The multi-objective formulation removes the need to hand-tune the detection/localization trade-off, and the same MOTPE protocol transfers to a ResNet-18 backbone, so the contribution is methodological rather than architecture-specific; the evaluation is fully reproducible, with per-seed results and extended tables archived openly. Set against this, the benefit of Pareto selection over simple scalar criteria is partial and concentrated in localization rather than uniform across metrics, and the reported accuracy is bounded to the in-domain FantasyID setting: zero-shot transfer to SIDTD is near chance and is not recovered by standard augmentation, so cross-domain deployment would require multi-source training. The nested protocol is also computationally heavy, though a single deployed model retrains in minutes.

Applying the identical MOTPE protocol to a ResNet-18 backbone yields comparable blind-test performance (Sec. 4.6), so the approach is not tied to one architecture. By contrast, zero-shot evaluation on SIDTD exposed a substantial cross-domain gap; closing it through multi-source training is a priority for future work, alongside forgery-artifact features (noise fingerprints, frequency anomalies [6]) and foundation model backbones [32] for richer pre-training.

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